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From Text to Forecasts: Bridging Modality Gap with Temporal Evolution Semantic Space

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Time-Series Forecasting

Broad real-world applications

- Time-series forecasting is widely used in finance, environment, and traffic.
- Unimodal forecasting methods rely only on numerical histories.
- In stationary scenarios, these methods have achieved excellent performance.



Finance
news and sentiment shocks



Environment
weather and external factors



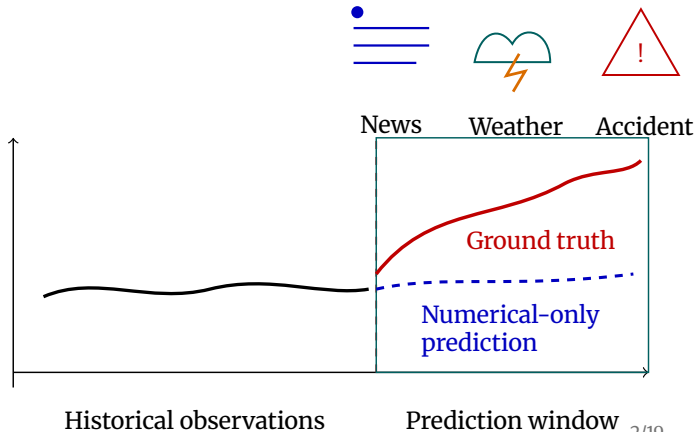
Traffic
accidents and demand shifts



Why Multimodal?

Event-driven non-stationarity

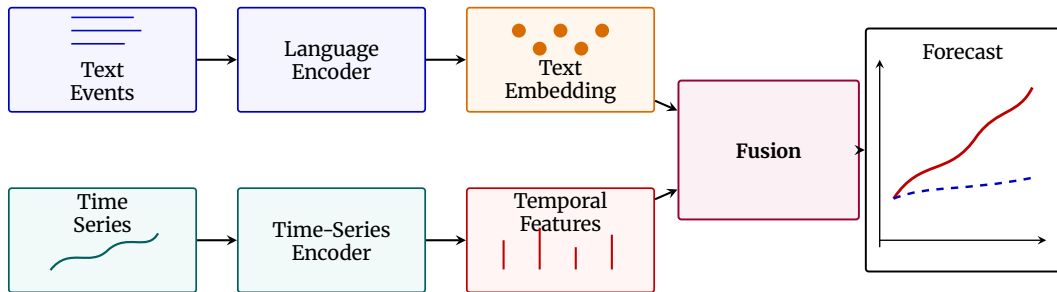
- However, real-world time series are often non-stationary: future statistical properties can shift substantially from the past.
- Such non-stationarity is often driven by external events.
- Under such shifts, numerical histories alone are insufficient.
- Text naturally encodes external events and provides additional information for forecasting.



Existing Methods

Recent work improves **non-stationary forecasting** by incorporating **textual information**, such as news and external events.

Most methods encode text and numerical histories separately, then fuse their representations via **cross-modal attention** (Jin et al., 2024; Wang et al., 2025; Wang et al., 2024).





The Modality Gap

Text often contains

- implicit, indirect temporal impacts
- vague, non-quantifiable expressions
- redundant context

**Modality
Gap**

Forecasting needs

- explicit information directly tied to future evolution
- precise, quantifiable signals

How do existing methods behave under this modality gap?



Diagnosing the Modality Gap

We diagnose how the **modality gap** affects forecasting performance.

Semi-synthetic data:

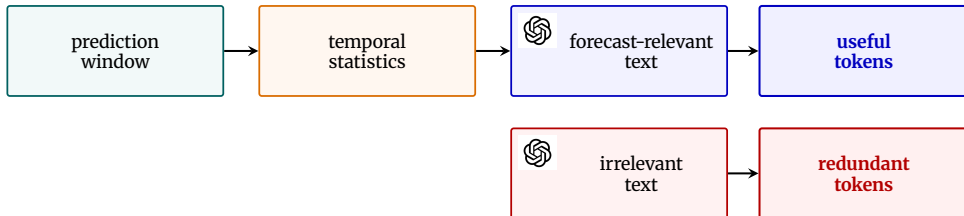
Why?

Real data lacks token-level labels, making it difficult to diagnose why existing methods fail. Fully synthetic data may distort temporal evolution.

How?

We use real time-series samples, generate text from each future window's statistics as **useful tokens**, and inject noisy text as **redundant tokens**.

How we construct the diagnostic text



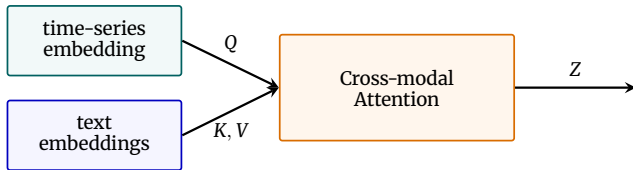


Attention Misalignment

Cross-modal attention

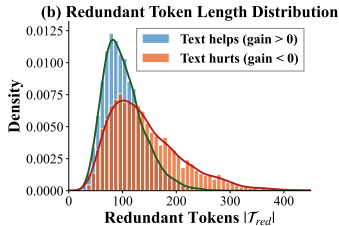
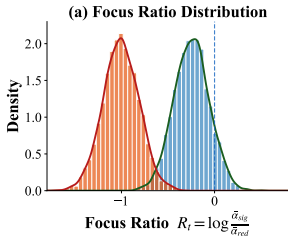
$$Q_X = XW_Q, \quad K_T = TW_K, \quad V_T = TW_V,$$

$$A = \text{softmax}\left(\frac{Q_X K_T^\top}{\sqrt{d}}\right), \quad Z = AV_T.$$



Findings

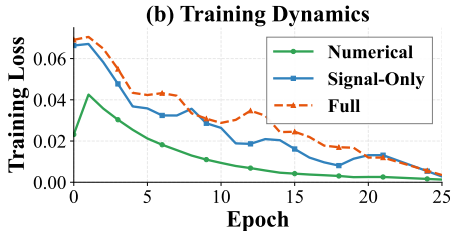
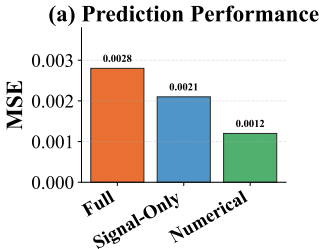
- More redundant tokens lead to worse forecasting performance.
- Even gain-positive samples still over-attend to redundant tokens.



Semantic Mismatch

Findings

- Even after removing all redundancy, the performance gain remains limited.
- Text is qualitative and ambiguous, making it hard to map directly into the numerical space.





Key Idea

Temporal Evolution Semantic Space (TESS)

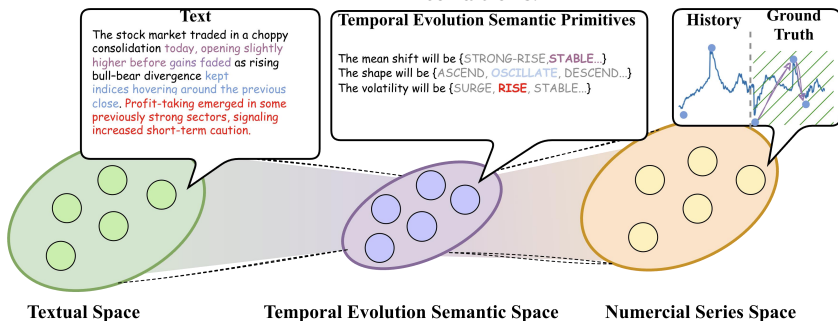
Role

A mediator for information exchange between textual descriptions and numerical time series.

Requirements

Evolution-relevant: directly tied to future temporal dynamics.

Quantifiable: grounded as numerical forecasting conditions.



Temporal Evolution Semantic Space

Following expert forecasting practice [1–5], we extract implications for temporal statistics instead of raw descriptions.



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Distribution Shift



distribution changes
strong-rise, stable,
strong-drop

Volatility Shift



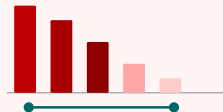
fluctuation becomes
stronger or calmer
surge, rise, stable,
fall, calm

Shape



forecast morphology
inside the horizon
ascend, descend, peak,
trough, oscillate

Lag & Decay

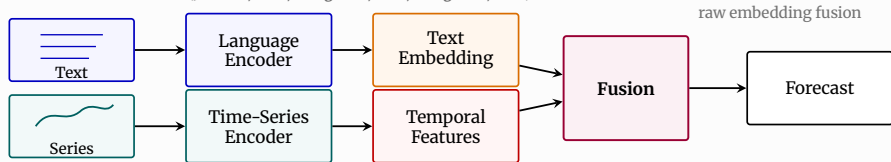


when impact starts and
how long it lasts
early-fade, early-persist,
late, diffuse

Comparison

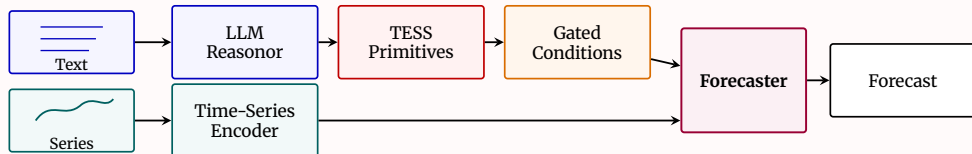


Previous methods (Jin et al., 2024; Wang et al., 2025; Wang et al., 2024)



TESS (ours)

semantic primitive mediation

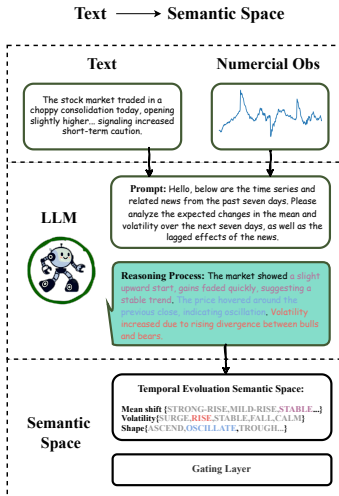




Stage 1: Text $\rightarrow$ TESS

Text to semantic primitives

- A frozen LLM reads text and numerical observations together.
- It extracts temporal evolution primitives in TESS.
- Confidence-aware gating suppresses unreliable primitive labels.



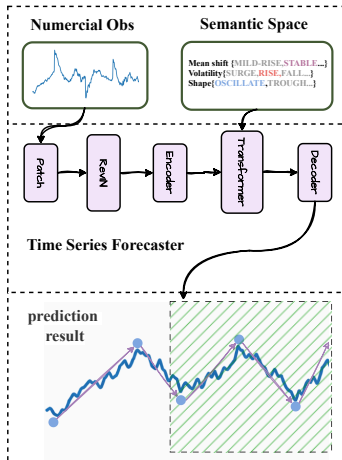


Stage 2: TESS $\rightarrow$ Forecast

Semantic primitives to numerical forecasts

- Gated primitives condition a PatchTST-based forecaster.
- The model fuses compact semantic labels with historical patches.
- The output is a numerical forecast grounded in temporal dynamics.

Semantic Space $\longrightarrow$ Numerical Space



Main Results



MSE comparison (lower is better)

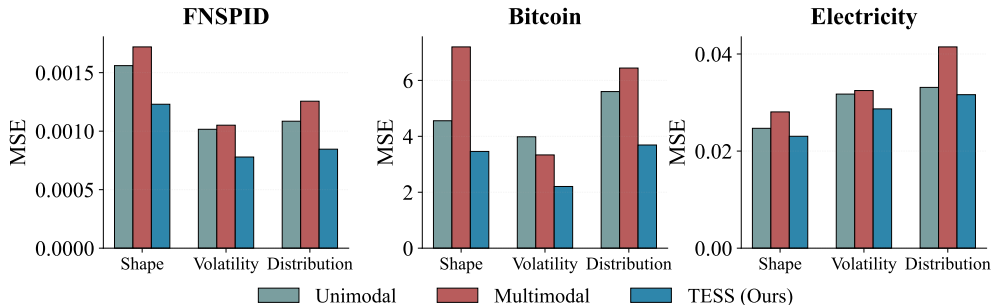
Model	Bitcoin	FNSPID	Electricity	Environment
Nonstationary	4.3420	0.0016	0.0256	0.3472
TimesNet	3.8229	0.0015	0.0242	0.3473
NewsForecasting	3.2036	0.0017	0.0254	0.3678
TESS (Ours)	2.2726	0.0012	0.0230	0.3534
<i>Gain vs. best</i>	+29.1%	+20.0%	+5.0%	−1.8%

- **Financial datasets:** substantial gains under pronounced event-driven non-stationarity.
- **General datasets:** best on Electricity and runner-up on Environment, showing stable predictive capability.
- **Overall:** up to **29.1%** MSE reduction over the strongest baseline.



Robust under non-stationary scenarios

Evaluation on three non-stationary patterns



- We extract three test subsets: **shape transition**, **volatility change**, and **distribution shift**.
- TESS consistently improves across all three non-stationary scenarios.
- FNSPID/Bitcoin: **21–52% lower MSE vs. multimodal**, **21–45% vs. unimodal** baselines.

Ablation Study

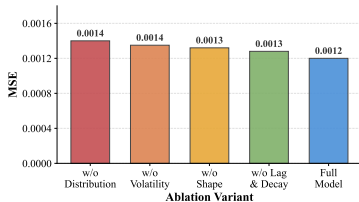


Module Ablation

MSE	w/o TESS	w/o Gate	Full
Bitcoin	4.2238	2.3556	2.2726
FNSPID	0.0018	0.0015	0.0012
Electricity	0.0298	0.0248	0.0230

Primitive Ablation

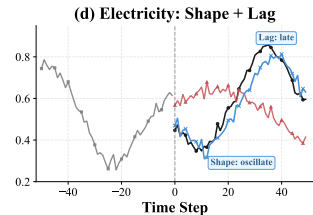
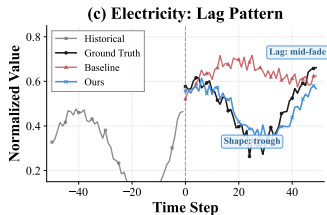
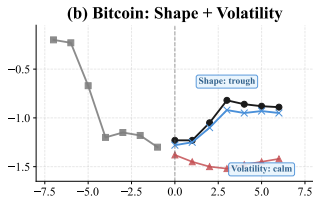
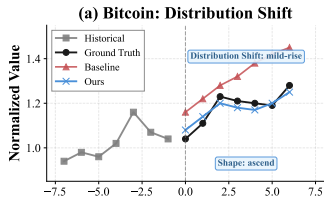
Primitive Ablation on FNSPID



Findings

- Removing TESS causes the largest degradation across datasets.
- Confidence-aware gating provides smaller but consistent gains.
- All four primitives are complementary.
- Distribution shift is the most influential primitive on FNSPID.

Case Study



Takeaway

- TESS captures event-induced temporal primitives: distribution shift, trend, shape, and lag.
- These primitives translate implicit text signals into forecasts that align closer to ground truth.
- Direct text fusion misses critical temporal patterns.

Future Work

1 **Adaptive, data-driven primitive learning**

Move beyond a fixed set of hand-designed primitives and learn task- or domain-specific temporal semantics from data, while preserving numerical verifiability and interpretability.

2 **Multivariate multimodal time-series forecasting**

Extend TESS to settings where multiple variables interact and external text may affect several channels through shared events, delayed effects, and cross-series dependencies.

Acknowledgements

Thank You!

Questions?

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